

what is difference between grinding and polishing in lens manufacturing?

In lens manufacturing, **grinding** and **polishing** are different steps.

Simple difference

Item	Grinding	Polishing
Main purpose	Make the lens shape	Make the surface smooth and optically clear
Removes material	More material	Very small amount
Surface result	Rough / matte surface	Smooth / transparent surface
Accuracy focus	Radius, thickness, diameter, curve shape	Surface roughness, clarity, optical quality
Tool type	Diamond wheel, abrasive tool	Polishing pad, pitch, polyurethane pad, slurry
Typical stage	Early / middle process	Final surface finishing process
Output	Correct geometry but not clear	Clear optical surface

1. Grinding

Grinding is the process to create the lens shape.

For example, if you need a spherical lens, aspheric lens, or plano lens, grinding removes glass material to form the required curve.

Grinding controls:

Controlled item	Meaning
Diameter	Lens outer size, for example 50 mm, 100 mm, 200 mm
Thickness	Center thickness / edge thickness
Radius of curvature	The curved surface shape
Basic form accuracy	How close the shape is to the target
Stock removal	Removing extra glass material

After grinding, the lens surface is usually **not transparent**. It may look cloudy or frosted because many tiny scratches remain.

2. Polishing

Polishing is the process to make the ground lens surface optically smooth.

After grinding, the surface has micro-scratches and subsurface damage. Polishing removes a very thin layer and makes the surface smooth enough for light to pass through clearly.

Polishing controls:

Controlled item	Meaning
Surface roughness	How smooth the surface is, for example $R_a \leq 2 \text{ nm}$
Scratch removal	Removes grinding marks
Transparency	Makes lens clear
Figure correction	Improves final optical shape
Surface quality	Final optical-grade surface

After polishing, the lens becomes **transparent and usable for optical systems**.

Easy example

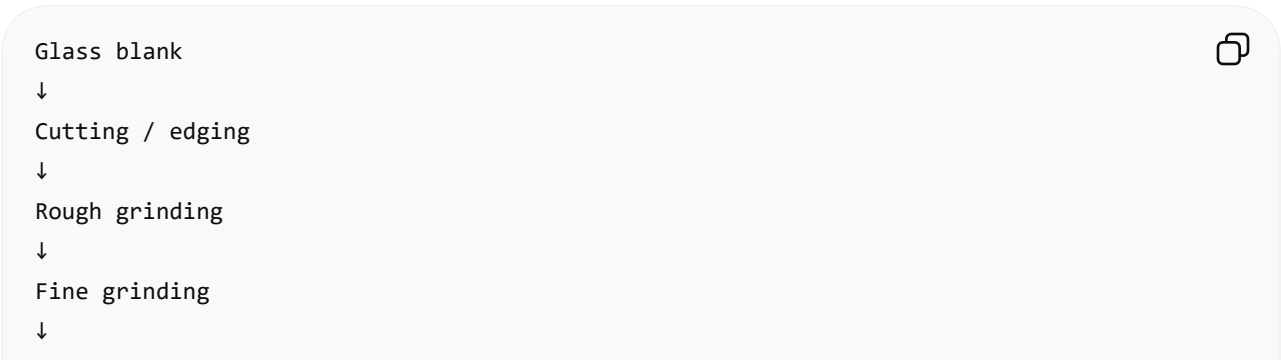
Think about making a glass lens like making a table surface:

Step	Similar example
Grinding	Cutting and shaping wood to the correct size
Polishing	Sanding and finishing the surface until it is smooth and shiny

Grinding gives the **shape**.

Polishing gives the **optical quality**.

Typical lens process flow



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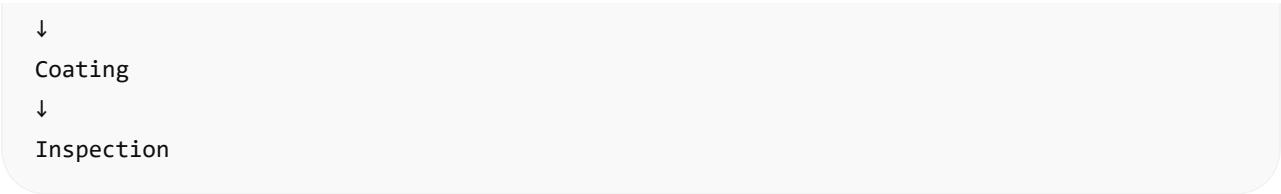
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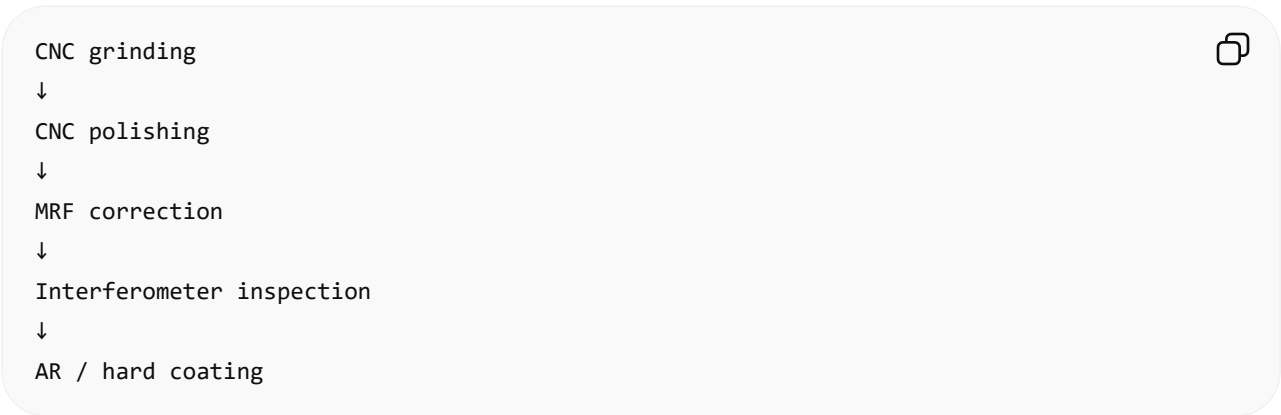
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For high-precision lenses, the flow may be:



Key point

For your lens specification like:

Diameter: 50–200 mm

Material: BK7 / fused silica / crown / flint glass

Surface quality: $Ra \leq 2$ nm

Lens type: spherical / aspheric / freeform / plano



You normally need both:

Grinding = make accurate lens geometry

Polishing = achieve optical smoothness and transparency



Grinding alone is not enough because the surface will be too rough for optical use.

Polishing alone is not enough because it cannot efficiently create the full lens shape from raw glass.

